

Feature

New Horizons has brought to light some interesting facts about our favourite dwarf planet



What's going on NASA?

New Horizons and Dawn both suffer glitches. Looks like a problematic July
<http://digit.in/glitchy>

Ticket to a new world

You don't need to travel to deep space to find a wormhole. They may be closer than you think

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Thanks to *Interstellar*, we all now know what wormholes are. The idea of portals helping us travel in space-time sounds fascinating and intriguing, but do these wormholes actually exist? On paper they certainly do. What's even more fascinating is that unlike the movie where the intrepid astronauts had to travel all the way to Saturn to find one, there's a possibility that wormholes could be discovered much closer to home. A group of NASA scientists are planning to prove the existence of such portals by looking for one around our very own planet Earth.

But first...

For those who came in late, you're probably wondering what wormholes are. Here's a quick lesson before we speak about the mission. "Einstein-Rosen bridges" are, as the name suggests, bridges that connect two different points in space and time. While studying the mathematical solution for black holes, Albert Einstein and American-Israeli physicist Nathan Rosen discovered that the solution extended itself to give another solution that described two black holes connected by a throat. This throat that acts as a bridge between two wormholes is called the "Einstein-Rosen bridge" or simply a "wormhole". They also found that these wormholes

are extremely unstable and will collapse really quickly.

Later though it was found that wormholes can, in fact, be stable and even exist for long durations if the conditions are right – specifically if a large amount of energy called "exotic matter" is present. This exotic matter will have negative energy and negative mass, which give it anti-gravity properties. This exotic matter will prevent a wormhole from collapsing.

How it all started

Before we move on to explain what this mission is all about, let's look at the phenomenon that forms the basis of this undertaking.

CMEs

The sun, as quiet as it seems, can get really restless and can start throwing tantrums.



CMEs striking earth's magnetosphere

This outburst can be as nonchalant as a burp or as dangerous as a million hydrogen bombs. These explosions are commonly known as "coronal mass ejections" or simply "CMEs".

During a CME, enormous bubbles of superheated gas – called plasma – are ejected from the sun. Over the course of several hours, one billion tons of material are lifted off the sun's surface and accelerated to speeds of one million miles per hour. The frequency of these emissions can vary from five times a day to once every five days.

These ejections can travel up to several million miles per hour and the high energy charged particles carried with it can prove to be very lethal. But, if these bursts are so dangerous and are striking our planet every second of every day, how are we still alive?

Magnetosphere

Due to its molten iron-nickel core, earth acts as a huge dipole magnet. It has helped us navigate our way around the planet for centuries now. But it has a much bigger role than that. "Magnetosphere" is the area in space that's controlled by a planet's magnetic field. The boundary where a planet's magnetic field stops dominating and the sun's magnetic field starts taking over is called the "magnetopause".